

‘Necessary in the highest degree—whatever that means’

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1 What is necessity *tout court*, for Kripke?

Textual evidence that whatever is necessity *tout court* is necessary in every way:

First, that characteristic theoretical identifications like ‘Heat is the motion of molecules’, are not contingent truths but necessary truths, and here of course I don’t mean just physically necessary, but necessary in the highest degree—whatever that means. (Physical necessity, *might* turn out to be necessity in the highest degree. But that’s a question which I don’t wish to prejudge. At least for this sort of example, it might be that when something’s physically necessary, it always is necessary *tout court*.) (Kripke 1972: 99, henceforth NN)

So if this consideration is right, it tends to show that such statements representing scientific discoveries about what this stuff is are not contingent truths but necessary truths in the strictest possible sense. It’s not just that it’s a scientific law, but of course we can imagine a world in which it would fail. (NN: 125)

If Kripke had been open to there being other forms of necessity, more demanding than necessity *tout court*, why would he have interpreted his opponents (e.g. those who equated ‘necessarily *P*’ with ‘“*P*” is analytic’) as speaking (falsely) about necessity *tout court*, rather than something else?

Objection: this interpretation would make Kripke’s claims about necessity *tout court* obviously false!

In the first place, truths of logic—truths of propositional logic and first order quantification theory, let us say—are necessary in the sense in question. Such truths are logically necessary in the narrow sense. But the sense of necessity in question—call it ‘broadly logical necessity’—is wider than this. Truths of set theory, arithmetic and mathematics generally are necessary in this sense, as are a host of homelier items such as

No one is taller than himself

Red is a colour

If a thing is red, then it is coloured

No numbers are human beings

and

No prime minister is a prime number

... So the sense of necessity in question is wider than that captured in first order logic. (Plantinga 1974: 1–2)

Response: although one could (foolishly) introduce a convention whereby ‘It is *L*-necessary that *P*’ expresses a truth exactly when *P* is a theorem of a given logic *L*, that would involve making an exception to the general compositional-semantic rules for complex expressions; so the truth and falsity of such sentences doesn’t tell us about a form of necessity expressed by ‘*L*-necessary’ (if ‘*L*-necessary’ expresses anything at all).

2 The absolute necessity of identity

A key Kripkean strategy for arguing for controversial (at the time) necessity claims is to appeal to the following principle (a theorem of a very minimal quantified modal logic):

$$\text{NI} \quad \forall xy. x = y \rightarrow \Box(x = y)$$

For example:

- (A) 1. Hesperus is Phosphorus
- 2. Therefore, it is necessary that Hesperus is Phosphorus (from NI, by universal instantiation).
- 3. Therefore, it is necessary that anyone who sees Hesperus sees Phosphorus (by uncontroversial modal logic).

Objection: superficially similar universal instantiations are clearly invalid when using definite descriptions.

Response: proper names, unlike typical definite descriptions, are rigid designators. Cash value of ‘*N* is a rigid designator’: $\forall x. x = N \rightarrow \Box(x = N)$ is true.

What motivates this? Scopelessness intuitions.

With these extra rigidity premises, we can do without UI into the modal operator:

- 1. $\forall xy. x = y \rightarrow \Box(x = y)$
- 2. $\forall x. x = \mathbf{h} \rightarrow \Box(x = \mathbf{h})$
- 3. $\forall y. y = \mathbf{p} \rightarrow \Box(y = \mathbf{p})$
- 4. $\mathbf{h} = \mathbf{p}$
- 5. $\exists xy. x = y \wedge x = \mathbf{h} \wedge y = \mathbf{p}$ by 4
- 6. $\exists xy. \Box(x = y) \wedge \Box(x = \mathbf{h}) \wedge \Box(y = \mathbf{p})$ by 5, 1, 2, 3
- 7. $\exists xy. \Box(x = y \wedge x = \mathbf{h} \wedge y = \mathbf{p})$ by 6
- 8. $\exists xy. \Box(\mathbf{h} = \mathbf{p})$ by 7
- 9. $\Box(\mathbf{h} = \mathbf{p})$ by 8

Objection: The intuitions only support rigidity/scopelessness of proper names for *ordinary* interpretations of modal words. But the use of ‘necessarily’ to mean absolute necessity is not ordinary.

Response: The rigidity of names with respect to familiar modalities is evidence that names have a different *type* of meaning from definite descriptions. The type of meaning that proper names have and ordinary (non-epistemic) modal operators have guarantee that any expression with the former type of meaning is scopeless with respect to any operator with the latter type of meaning, including absolute necessity.

3 Beyond proper names

The strategy is supposed to generalize beyond names:

Theoretical identities, according to the conception I advocate, are generally identities involving two rigid designators and therefore are examples of the necessary a posteriori. (NN: 140).

What are these ‘theoretical identities’?

Philosophers have, as I’ve said, been very interested in statements expressing theoretical identifications; among them, that light is a stream of photons, that water is H₂O, that lightning is an electrical discharge, that gold is the element with the atomic number 79. (NN: 116)

Let’s consider the statement ‘Light is a stream of photons’ or ‘Heat is the motion of molecules’. By referring to light, of course, I mean something which we have some of in this room. (NN: 98)

Similarly for many other such identifications, say, that lightning is electricity. Flashes of lightning are flashes of electricity. Lightning is an electrical discharge. (NN: 132)

Objection: these sentences are just not identities (Soames 2002: ch. 9)! Some further “essentialist” premise would be needed to derive any modal conclusions from them. And maybe this premise isn’t true for *absolute* necessity (Kment 2014; Salmón 2024).

My view: most of these examples do have readings as plausible identities, which imply their own necessity for the same reason ‘Hesperus is Phosphorus’ does.

But let’s first consider some more straightforward examples:

- (B) 1. The property of being golden is the property of being made of atoms with 79 protons.
 2. Therefore, it is necessary that the property of being golden is the property of being made of atoms with 79 protons.
 3. Therefore, it is necessary that every golden thing is made of with 79 protons.

- (C) 1. The property of being a tadpole is the property of being a juvenile frog.
 2. Therefore, it is necessary that the property of being a tadpole is the property of being a juvenile frog.
 3. Therefore, it is necessary that every tadpole is a juvenile frog.
 (D) 1. To be golden is to be made of atoms with 79 protons.
 2. Therefore, it is necessary that to be golden is to be made of atoms with 79 protons.
 3. Therefore, it is necessary that every golden thing is made of atoms with 79 protons.
 (E) 1. To be a tadpole is to be a juvenile frog.
 2. Therefore, it is necessary that to be a tadpole is to be a juvenile frog.
 3. Therefore, it is necessary that every tadpole is a juvenile frog.

(‘To be *F* is to be *G*’ also has other readings. I will always mean the one suggested by ‘to be *F* IS to be *G*’, or ‘to be *F* just is to be *G*’, equivalent to ‘to be *F* and to be *G* are one and the same’.)

I note in passing that Kripke has odd qualms about the word ‘property’:

Perhaps some ‘general’ names (‘foolish’, ‘fat’, ‘yellow’) express properties. In a significant sense, such general names as ‘cow’ and ‘tiger’ do not, unless *being a cow* counts trivially as a property. (NN: 127–8)

The type of property identity used in science seems to be associated with necessity, not with a prioricity, or analyticity: For all bodies *x* and *y*, *x* is hotter than *y* if and only if *x* has higher mean molecular kinetic energy than *y*. Here the coextensiveness of the predicates is necessary, but not a priori. The philosophical notion of attribute, on the other hand, seems to demand a priori (and analytic) coextensiveness as well as necessary coextensiveness. (NN: 138)

4 Denying absolute rigidity

A simplified analogue of Kment’s view: when ‘*F*’ is syntactically complex, ‘the property of being *F*’ is an absolutely nonrigid (though metaphysically rigid) description. E.g.: ‘the property of being *F* and *G*’ is equivalent to ‘the conjunction of the property of being *F* and the property of being *G*’. Which properties are the conjunctions of which is an absolutely contingent matter.

Response 1: The case for rigidity/scopelessness of expressions of the form ‘the property of being *F*’ w.r.t. modal operators looks similarly good to the case for names.

Response 2: If I’m right that ‘to be *F* is to be *G*’ is a *higher order* identity in which ‘[to] be *F*’ still has its ordinary predicate meaning (as in ‘She is about to be surprised’), the analogous response doesn’t even make sense for them.

5 Denying the identities

[ADD Caie as another target here] Soames (2002: tentatively) and Salmón (2024: confidently) maintain that although property identities like (B1) and (C1) would indeed have those modal consequences, they are false, since:

- (i) They imply false attributions of a priori knowability, like ‘It is knowable a priori that everything golden is composed of atoms containing 79 protons’.
- (ii) They imply false generalizations about attitudes and speech reports, like ‘It’s impossible to believe/claim that something is golden believes/claims but not that it is composed of atoms containing 79 protons’.

Salmón (2024) seems to suggest that this carries over to sentences with infinitives, like (D1) and (E1).

Responses:

1. !?!

2. Note that these authors accept ‘It is knowable a priori that Hesperus is Phosphorus’ and ‘Everyone who believes/claims that Hesperus is bright believes/claims that Phosphorus is bright’ as (unequivocally) true, so they are in a more delicate position than others might be in insisting on falsehood in the cases at hand. Meanwhile, anyone who *didn’t* accept this view — e.g. Kripke in 1972 — should definitely not think that *property* identities have these attitudinal consequences that identities involving names do not have!

3. Someone might be tempted to back up argument (ii) with some generalization like ‘In order to claim that something is composed of atoms with 79 protons, one must produce an expression-token that expresses *being a proton*’, or ‘In order to believe that something is composed of atoms with 79 protons, one must grasp [or: have a Mentalese term for] *being a proton*’. But examples involving predicates introduced by stipulative definition (e.g. in mathematics), but then picked up by speakers who don’t know the definition, put pressure on such generalizations (for proponents of substitutivity). For it would be *really* implausible to deny the truth of ‘The property of being a semigroup is the property of being an associative magma’ or ‘To be a semigroup is to be an associative magma’.

4. I rather think it *is* knowable a priori that everything golden is composed of atoms with 79 protons. Imagine aliens who have never seen gold and have no word for ‘gold’, but have learnt a lot about us using their scanners, including that when we get married we often exchange rings composed of atoms with 79 protons. Don’t they know that when we get married we often exchange golden rings? If so, don’t they also know that everything golden is composed of atoms with 79 protons? And if so, don’t they know this a priori?

5. Even if we accept that the target identities imply substitutivity in attitude reports, many of the implausible consequences of this can be blocked by appeal to context sensitivity in speech and attitude verbs. Maybe when we express a property with a complex predicate, that pushes us towards a correspondingly demanding reading

of ‘believe’, ‘claim’, etc. (Goodman and Lederman 2021)

6. But actually, that’s not the view of speech and attitude verbs I like. In my view, these verbs semantically aberrant: they are sensitive to the syntactic structure of their argument.

A toy model:

‘*x* said that *y* is a tadpole’ $\approx$ ‘*x* produced an utterance consisting in an expression referring to *y* combined (as an argument) with an expression referring to *being a tadpole*’.

‘*x* said that *y* is a juvenile frog’ $\approx$ ‘*x* produced an utterance consisting in an expression referring to *y* combined as an argument with a compound conjunctive expression consisting of an expression referring to *being juvenile* and an expression referring to *being a frog*’.

Each occurrence of ‘refer’ on the RHS of these equivalences involves a context-sensitive ‘in an appropriate way’ restriction; these can be, and often are, resolved non-uniformly.

(One could give a similar LOT-based account of ‘believe’; the general template can also be implemented in other ways.)

Why believe such a view, rather than just appealing to context-sensitivity in speech and attitude verbs?

- ‘Ellipsis’ argument: Not only ‘*X* believes that Hesperus is bright and does not believe that Phosphorus is bright’, but also ‘*X* believes that Hesperus is bright but not that Phosphorus is bright’, can be used (literally) to say something true. But with only one occurrence of ‘believe’, we can’t appeal to differential resolution of context-sensitivity.
- Comparative argument: ‘*X* is more confident that Hesperus is bright than that Phosphorus is’ can be used (literally) to say something true. But ‘more confident’ can only express irreflexive relations.

6 An ambiguity in ‘proposition’ talk

On this view, quantification over ‘things said/believed’ does not work like quantification into sentence position.

- (1) Although Cal believes there are tadpoles in the tank, unlike Sue he doesn’t believe there are juvenile frogs in the tank. So, he doesn’t believe everything she believes.

We can use ‘thoughts’ for what’s quantified over here. On the toy theory, ‘the thought that *y* is a tadpole’ could be equated with ‘the property of being a sentence consisting of an expression referring (appropriately) to *y* combined with an expression referring to *being a tadpole*’, and ‘*x* believes *t*’ would mean ‘there is an instance of *t* in *x*’s belief box’.

Philosophers use ‘proposition’ in two ways: sometimes for thoughts, sometimes for ‘states of affairs’ — the 0-adic analogue of properties: something formalized by

quantification into sentence position.

7 Back to Kripke's theoretical identifications

Claim: 'Fs are Gs' (e.g. 'Flashes of lightning are flashes of electricity') has a reading equivalent to 'To be F is to be G'. For exmple, (2) can mean the same as (E1) ('To be a tadpole is to be juvenile frog'):

(2) Tadpoles are juvenile frogs.

Of course, sentences with bare plural subjects also have "characterizing" readings. On a characterizing reading, (3) is true as well as (2):

(3) Tadpoles are frogs.

But given kind-predicating uses like

(4) Tadpoles are the most popular kind of animal to keep in kindergarten classrooms.

it would be mysterious if the kind-predicating use was not available (or trivial) for sentences like (2).

Scopelessness intuitions are in full force:

(5) ?? Although tigers could have been widespread, it couldn't have happened that tigers were widespread.

One might think that the kind-predicating uses (including the identities) require replacing the predicate meaning of 'tadpoles' (as in 'Those three things that swam under the rock are tadpoles') with some surrogate individual. Although that view wouldn't be a disaster for me, I think it's wrong:

(6) The three animals in this cage are axolotls, which are near extinction in the wild.

Everything just said about bare plural nominals also applies to bare mass nominals (including abstract nominals):

(7) Kumis is fermented mare's milk.

(8) Deuterium is hydrogen in which every atom has a neutron as well as a

proton

(9) Wisdom is knowledge of right and wrong.

(10) Heat is molecular motion.

Can this account carry over to sentences with indefinites, like the following?

(11) Light is a stream of photons.

(12) Lightning is an electrical discharge.

(13) Shandy is a 50/50 combination of beer and lemonade.

Maybe! It's well known that indefinite NPs can function as predicates (Strawson 1950: 320; Graff 2001), as in

(14) This candidate is a wonderful philosopher and worth taking very seriously if you have an opening

So it would make *semantic* sense for (11)–(13) to have readings as higher-order identities, equivalent to (15)–(17):

(15) To be light is to be a stream of photons.

(16) To be lightning is to be an electrical discharge.

(17) To be shandy is to be a 50/50 combination of beer and lemonade.

Objection: Even if (11) and (12) have readings equivalent to (15) and (16), these are so obviously false that Kripke couldn't have intended them. Streams of X-ray photons aren't light, and sparks caused by carpets aren't lightning!

Reply: actually, 'light' actually does have a broad use that covers all electromagnetic radiation:

(18) X-rays are a form of light, but much more energetic than the light detected by our eyes (https://www-xray.ast.cam.ac.uk/xray_introduction/Xrays.html)

And I *think* 'lightning' has a broad use that also applies to man-made sparks:

(19) You can create mini lightning bolts with a Van de Graaff generator using a discharge sphere (https://shop.wf-education.com/media/wysiwyg/downloads/van_der_graaff_1_2022_2_.pdf)

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